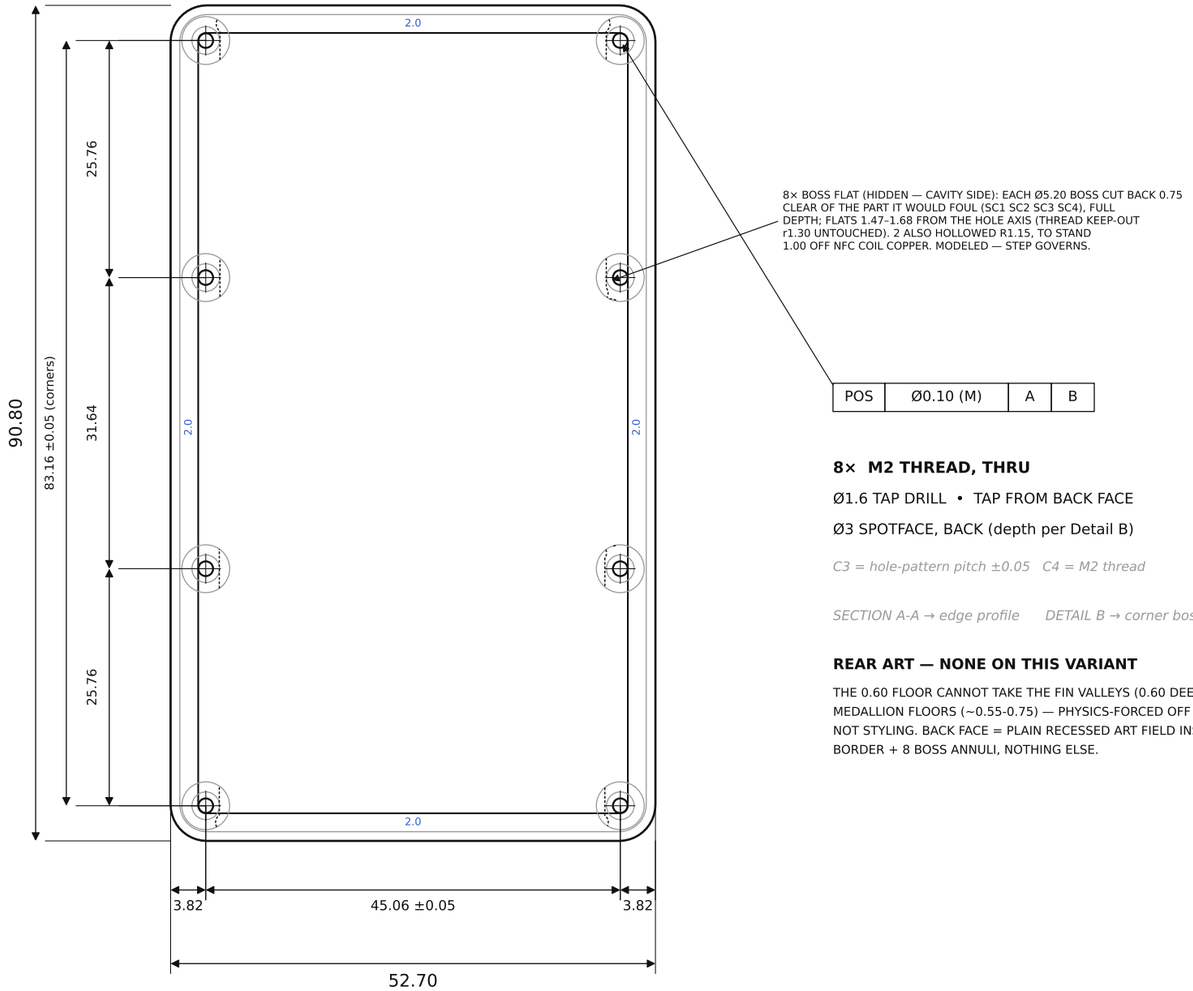
BACK FACE (outer) SCALE 1.5:1



8× BOSS FLAT (HIDDEN — CAVITY SIDE): EACH Ø5.20 BOSS CUT BACK 0.75 CLEAR OF THE PART IT WOULD FOUL (SC1 SC2 SC3 SC4), FULL DEPTH; FLATS 1.47–1.68 FROM THE HOLE AXIS (THREAD KEEP-OUT r1.30 UNTOUCHED). 2 ALSO HOLLOWED R1.15, TO STAND 1.00 OFF NFC COIL COPPER. MODELED — STEP GOVERNS.

POS	Ø0.10 (M)	A	B

8× M2 THREAD, THRU

Ø1.6 TAP DRILL • TAP FROM BACK FACE

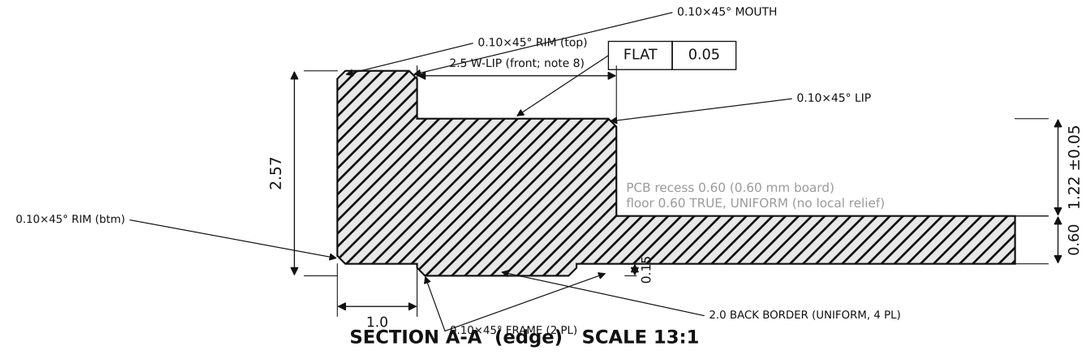
Ø3 SPOTFACE, BACK (depth per Detail B)

C3 = hole-pattern pitch ±0.05   C4 = M2 thread

SECTION A-A → edge profile   DETAIL B → corner boss

REAR ART — NONE ON THIS VARIANT

THE 0.60 FLOOR CANNOT TAKE THE FIN VALLEYS (0.60 DEEP) OR THE MEDALLION FLOORS (~0.55-0.75) — PHYSICS-FORCED OFF IN fit_rules.VARIANTS, NOT STYLING. BACK FACE = PLAIN RECESSED ART FIELD INSIDE THE 2.0 BORDER + 8 BOSS ANNULI, NOTHING ELSE.



SECTION A-A (edge) SCALE 13:1

C1 = cavity depth   C2 = PCB-rest-plane flatness   (back face down; PCB drops in from top)

DETAIL B (corner boss) 13:1

CRITICAL DIMENSIONS — C1 & C3 TOLERANCED ±0.05 (MARKED ON VIEWS); ALL OTHER FEATURES PER ISO 2768-1 (MEDIUM)

C1	Cavity depth (boss-top plane → cavity floor)	1.22 ±0.05
C2	PCB-rest plane flatness (lip + 8 bosses)	FLAT 0.05
C3	8× mounting holes (x 45.06; y rows 2.87/28.63/60.27/86.03)	±0.05 (linear)
C4	Mounting holes — tapped, thru, from back	M2 / Ø1.6 drill

GENERAL TOLERANCES (UNLESS OTHERWISE NOTED, mm)

LINEAR & HOLE-POSITION DIMS	**ISO 2768-m**
CAVITY DEPTH C1 (SECTION A-A)	**1.22 ±0.05**
HOLE-PATTERN PITCH C3 (8 HOLES)	**x 45.06 / y 25.76-31.64-25.76 ±0.05**
PCB-REST FLATNESS C2	**FLAT 0.05**

SOLAR-GLOW DRH — Ti BACK-SHELL LITE (0.60 FLOOR, THINNED)

DWG solar-glow-drh-shell-lite-Ti	REV lite
MATERIAL Ti Gr5 (TC4)	FINISH BEAD-BLAST MATTE
UNITS mm	SCALE AS NOTED
BBOX 52.70 x 90.80 x 2.57 3rd-angle	SHEET 1/1

NOTES

1. MATERIAL: TITANIUM Gr5 (TC4) = Ti-6Al-4V GRADE 5. [Ti Gr5 (thinned floor; the 0.60 is the shop-minimum conversation)]
2. FINISH: BEAD-BLAST MATTE, ALL OVER. NO MARKING SERVICE. NO REAR ART — THE 0.60 FLOOR CANNOT TAKE THE FIN VALLEYS (0.60 DEEP) OR THE MEDALLION FLOORS (~0.55-0.75); BOTH PHYSICS-FORCED OFF (fit_rules.VARIANTS).
3. GENERAL TOLERANCE PER ISO 2768-1 (MEDIUM). C1-C4 TOLERANCED AS LISTED; C1 & C3 = ±0.05. DATUMS: A = LEFT EDGE, B = BOTTOM EDGE, C = PCB-REST PLANE.
4. BRACE REGISTRATION: THE RESIN BRACE (1 PIECE, ITS FOOTPRINT COMPUTED FROM THE BOARD) REGISTERS TO THIS SHELL BY FITMENT - ITS OUTLINE, 0.05 OFF THE CAVITY WALLS WHERE IT REACHES THEM, + THE COMPONENT POCKETS + THE BOARD IN ITS RECESS (RECESS 50.70 x 88.80 FOR A 50.80 x 88.90 BOARD: 0.05 INTERFERENCE PER FLAT AS MODELLED - A PRESS FIT BY DESIGN). NO LOCATOR PILLARS; THE CAVITY FLOOR IS ONE FLAT FACE AT 0.60.
5. EDGE BREAKS - ALL 0.10x45°, FELT NOT SEEN. MODELED, CALLED OUT ON SEC A-A / DETAIL B: RIM = outer rim (top & bottom); MOUTH = recess mouth (around PCB); FRAME = proud back-frame bottom edges; BOSS = boss + spotface bottom edges (Detail B). LIP = inner (cavity-side) lip edge: BREAK EVERY LIP EDGE ON THE PART, WHETHER OR NOT THE MODEL SHOWS IT. BREAK ALL OTHER EXPOSED EDGES 0.10x45°. CONCAVE JUNCTION CORNERS LEFT SHARP IN THE MODEL: FINISH WITH A Ø2.0 mm TOOL = R1.0 AS-MILLED; ALL CLEARANCES ALLOW FOR IT (BOSS-CORNER Ø2.0 BORES MODELED). SMALLER IS FINE, LARGER IS NOT.
6. FLOOR IS A TRUE 0.60, UNIFORM ACROSS THE CAVITY, BELOW THE ~1.0 Ti MIN-WALL GUIDANCE - THE 0.60 IS THE SHOP-MINIMUM CONVERSATION (fit_rules.VARIANTS); CONFIRM THE SHOP HOLDS IT BEFORE QUOTING. CAVITY 1.22 ±0.05 -> 1.17 WORST-CASE, MINUS THIN CAP 1.00 MAX (PROVISIONAL — part_heights.SUPERCAP_H_THIN) = 0.17 NON-CONTACT; THE BRACE + SOLAR-CELL SANDWICHES CARRY THE BOARD.
7. NO U7 RELIEF POCKET. U7 IS THE 0.90 mm MB85RC512TY DFN-8, WELL UNDER THE 1.22 CAVITY, SO THE CAVITY FLOOR IS ONE FLAT FACE AT 0.60. (A POCKET EXISTED FOR A 1.75 SOIC-8 THE v4 BOARD DOES NOT CARRY.)
8. FRONT SUPPORT LIP IS CUT PER BAND TO CLEAR B-SIDE PARTS (SECTION A-A: THE WEST LIP AT FULL WIDTH). WIDTHS FROM THE BOARD EDGE, POSITIONS IN BOARD x/y, LIMITER IN BRACKETS: WEST 2.50; 2.32 y44.5–47.5 (R16), 2.25 y47.5–50.5 (C24). EAST 1.25 FULL LENGTH (NFC COIL, JP1, TP1). SOUTH 2.00; NONE x3.0–23.5 (SC1), x27.0–47.5 (SC2). NORTH 2.00; NONE x3.5–23.5 (SC3), x27.5–47.5 (SC4). A BAND STOPS 0.30 SHORT OF ITS PART, AND 1.00 SHORT OF NFC COIL COPPER (GROUNDED METAL OVER THE COIL WOULD DETUNE IT). STEPS BLEND AT R1.00. THE EXTERIOR BACK BORDER (PLAN) IS INDEPENDENT AND UNIFORM 2.0 ON ALL 4 SIDES.
9. NO INTERNAL RIBS OR SUPPORT POSTS: THE RESIN BRACE (SEPARATE PART) CARRIES CENTER SUPPORT. A PCB LAYOUT CHANGE = A BRACE REPRINT, NOT A SHELL RE-MACHINE.
10. PCB RECESS = 0.60 DEEP, RECEIVES THE 0.60 mm BOARD (RECESS 50.70 x 88.80 FOR A 50.80 x 88.90 BOARD: 0.05 INTERFERENCE PER FLAT AS MODELLED - A PRESS FIT BY DESIGN). 3D STEP GOVERNS ALL GEOMETRY; STL NOT FOR CNC.
11. PART = BACK-SHELL ONLY. PCB, RESIN BRACE, AND 8× M2×2 BRASS SLOTTED-CHEESE SCREWS (HEAD Ø3.8) SUPPLIED SEPARATELY.